THQ #5

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Question 1

Objective Function:

$$\textbf{min} \ -2x_1 + 5x_2 + 9x_3$$

Constraints:

- $2x_2 + 5x_3 \geq 1$
- $3x_1 - x_2 - x_3 \leq 6$
- $2x_1 - 4x_2 + x_3 = 3$
- $x_1, x_2, x_3 \geq 0$ **(non-negative)**

From the initial tableau, the extra columns are:

- x_4: surplus for $2x_2 + 5x_3 \geq 1$
- x_5: slack for $3x_1 - x_2 - x_3 \leq 6$
- x_6: artificial for the $\geq$ constraint
- x_7: artificial for the equality

Initial basic variables: x_6, x_5, x_7.

Given Tableaux:

	x_1	x_2	x_3	x_4	x_5	x_6	x_7	
x_6	0	2	5	-1	0	1	0	1
x_5	3	-1	-1	0	1	0	0	6
x_7	2	-4	1	0	0	0	1	3
	-2	5	9	0	0	0	0	0
x_3	0	0	1	$-\frac{1}{6}$	$-\frac{1}{15}$	$\frac{1}{6}$	$\frac{1}{10}$	$\frac{1}{15}$
x_2	0	1	0	$-\frac{1}{12}$	$\frac{1}{6}$	$\frac{1}{12}$	$-\frac{1}{4}$	$\frac{1}{3}$
x_1	1	0	0	$-\frac{1}{12}$	$\frac{11}{30}$	$\frac{1}{12}$	$-\frac{1}{20}$	$\frac{32}{15}$
	0	0	0	$\frac{7}{4}$	$\frac{1}{2}$	$-\frac{7}{4}$	$\frac{1}{4}$	2

Dual:

The primal is a minimization, so the dual is a maximization. The objective coefficients are the primal right-hand sides, 1, 6, and 3. Since $x_1, x_2, x_3 \geq 0$, each dual constraint is $\leq$.

Objective Function:

$$\textbf{max} \;\; y_1 + 6y_2 + 3y_3$$

Constraints:

- $3y_2 + 2y_3 \leq -2$
- $2y_1 - y_2 - 4y_3 \leq 5$
- $5y_1 - y_2 + y_3 \leq 9$
- $y_1 \geq 0$ because the first primal constraint is $\geq$
- $y_2 \leq 0$ because the second primal constraint is $\leq$
- y_3 unrestricted because the third primal constraint is an equality

Optimal value (primal and dual):

The final tableau is optimal, so strong duality applies: the primal minimum and the dual maximum are equal. The bottom-right entry is 2, and that entry is the negative of the objective value, so

$$z^* = -2$$

for both the primal and the dual.

Optimal primal solution:

From the final tableau, the basic variables are x_1, x_2, and x_3. Their values are the right-hand side of that tableau, and every nonbasic variable is 0:

$$(x_1, x_2, x_3) = \left(\frac{32}{15}, \frac{1}{3}, \frac{1}{15}\right).$$

Plugging this point into the primal objective confirms the value above:

$$-2\left(\frac{32}{15}\right) + 5\left(\frac{1}{3}\right) + 9\left(\frac{1}{15}\right) = -\frac{64}{15} + \frac{25}{15} + \frac{9}{15} = -2.$$

Optimal dual solution:

The optimal dual point is read from the bottom row of the final tableau, under the columns that were the identity in the initial tableau. Those are the slack and artificial columns, one per primal constraint. Each bottom-row entry there is the negative of the corresponding dual variable.

- First constraint, artificial x_6: bottom entry $-\frac{7}{4}$, so $y_1 = -\left(-\frac{7}{4}\right) = \frac{7}{4}$.
- Second constraint, slack x_5: bottom entry $\frac{1}{2}$, so $y_2 = -\frac{1}{2}$.
- Third constraint, artificial x_7: bottom entry $\frac{1}{4}$, so $y_3 = -\frac{1}{4}$.

$$(y_1, y_2, y_3) = \left(\frac{7}{4}, -\frac{1}{2}, -\frac{1}{4}\right).$$

These signs match the dual restrictions: $y_1 \geq 0$, $y_2 \leq 0$, and y_3 unrestricted. The dual objective agrees with the shared optimal value:

$$\frac{7}{4} + 6\left(-\frac{1}{2}\right) + 3\left(-\frac{1}{4}\right) = \frac{7}{4} - 3 - \frac{3}{4} = -2.$$

Question 2

Objective Function:

$$\textbf{max} \;\; 10x_1 - 12x_2 + 11x_3$$

Constraints:

- $6x_1 - 7x_2 + 8x_3 = 90$
- $-x_1 + 3x_3 \geq 42$
- $x_1, x_2, x_3 \geq 0$ **(non-negative)**

The objective row in the tableaux is the minimized form of this problem: minimizing $-10x_1 + 12x_2 - 11x_3$.

From the initial tableau, the extra columns are:

- x_4: surplus for $-x_1 + 3x_3 \geq 42$
- x_5: artificial for the equality
- x_6: artificial for the $\geq$ constraint

Initial basic variables: x_5, x_6.

Given Tableaux:

	x_1	x_2	x_3	x_4	x_5	x_6	
x_5	6	−7	8	0	1	0	90
x_6	−1	0	3	−1	0	1	42
	−10	12	−11	0	0	0	0
x_3	$-\frac{1}{3}$	0	1	$-\frac{1}{3}$	0	$\frac{1}{3}$	14
x_2	$-\frac{26}{21}$	1	0	$-\frac{8}{21}$	$-\frac{1}{7}$	$\frac{8}{21}$	$\frac{22}{7}$
	$\frac{25}{21}$	0	0	$\frac{19}{21}$	$\frac{12}{7}$	$-\frac{19}{21}$	$\frac{814}{7}$

Dual:

The primal is a maximization, so the dual is a minimization. The objective coefficients are the primal right-hand sides, 90 and 42. Since $x_1, x_2, x_3 \geq 0$, each dual constraint is $\geq$. There are two primal constraints, so the dual has only y_1 and y_2.

Objective Function:

$$\textbf{min}\ \ 90y_1 + 42y_2$$

Constraints:

- $6y_1 - y_2 \geq 10$
- $-7y_1 \geq -12$
- $8y_1 + 3y_2 \geq 11$
- y_1 unrestricted because the first primal constraint is an equality
- $y_2 \leq 0$ because the second primal constraint is $\geq$

Optimal value (primal and dual):

The final tableau is optimal, so strong duality applies: the primal maximum and the dual minimum are equal. The bottom-right entry of the final tableau is the optimal value of the original maximize problem, so

$$z^* = \frac{814}{7}$$

for both the primal and the dual.

Optimal primal solution:

From the final tableau, x_2 and x_3 are basic and x_1 is nonbasic, so

$$(x_1, x_2, x_3) = \left(0, \frac{22}{7}, 14\right).$$

Optimal dual solution:

The optimal dual point is read from the bottom row of the final tableau, under the artificial columns that were the identity in the initial tableau. The tableau minimizes the negative of the original objective, so those entries are copied as written. No extra sign flip.

- First constraint, artificial x_5: bottom entry $\frac{12}{7}$, so $y_1 = \frac{12}{7}$.
- Second constraint, artificial x_6: bottom entry $-\frac{19}{21}$, so $y_2 = -\frac{19}{21}$.

$$(y_1, y_2) = \left(\frac{12}{7}, -\frac{19}{21}\right).$$

These signs match the dual restrictions: y_1 is unrestricted and $y_2 \leq 0$. The dual objective agrees with the shared optimal value:

$$90\left(\frac{12}{7}\right) + 42\left(-\frac{19}{21}\right) = \frac{1080}{7} - 38 = \frac{814}{7}.$$